UNIVERSITI TEKNOLOGI MARA
FINAL ASSESSMENT

COURSE	:	**NETWORK PROGRAMMING**
COURSE CODE	:	**ITT440**
EXAMINATION	:	**JULY 2025**
TIME	:	**3 HOURS**

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of **FOUR (4)** questions.

2. Answer **ALL** questions in an original source code format. Screenshot the results and save them in PNG image format. Each question may require multiple screenshots.

3. Please use the template from https://github.com/addff/2503-ITT440 GitHub repository. Penalties will be applied to students who do not use this template.

4. All answers must be uploaded to this link. Please zip the file using 7-Zip (.7z). ***https://ctncloud2024.ictff.com.my/mod/assign/view.php?id=414***

5. Kindly send all the provided responses to the email address shahadan@fskm.uitm.edu.my as a backup. Use the title **'2503-ITT40 Backup Final Lab Test (SBA) - [Student Full Name] – [GROUP]**'. Don't zip your file for email.

6. Please ensure that the backup answers are submitted no later than 15 minutes after the exam time is over.

   i) If there are technical problems, an additional 45 minutes will be given to complete the submission.
   ii) After this extended period, the lecturer reserves the right to reject the student's submission.
   iii) Any technical problems must be reported to the lecturer via SMS or Telegram at 019-5491911 within 1 hour after the exam time is over.

7. Answer **ALL** questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages

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QUESTION 1

Refer to the script below and answer all questions.

```
#!/bin/bash
timestamp=$(TZ=Asia/Kuala_Lumpur date)
mkdir -p 2503-ITT440_Final_Assessment/
read -p "Enter Your Shortname (one word only): " nama_singkatan
while true; do
 echo "Pilih bahagian untuk dijalankan:"
 echo "1. Section 1"
 echo "2. Section 2"
 echo "3. Section 3"
 echo "4. Exit"
 read -p "Masukkan pilihan anda (1/2/3/4): " pilihan
 case $pilihan in
 1)
# Section 1
  read -p "Enter Your Full Name (CAPITAL LETTER): " nama
  read -p "Enter Your Student ID: " nombor_pelajar
  cd 2503-ITT440_Final_Assessment/
  folder_name="${nama_singkatan}_q1"
  mkdir "$folder_name"
  cd "$folder_name"
  echo "// Name: $nama" > "2503-ITT440_${nama_singkatan}_q1.c"
  echo "// Student ID: $nombor_pelajar" >> "2503-ITT440_${nama_singkatan}_q1.c"
  echo "// Shortname: $nama_singkatan" >> "2503-ITT440_${nama_singkatan}_q1.c"
  echo "// Date: $timestamp" >> "2503-ITT440_${nama_singkatan}_q1.c"
  echo "// Q1 (FORK) 2503-ITT440 FA" >> "2503-ITT440_${nama_singkatan}_q1.c"
  echo "Maklumat telah disimpan dalam fail 2503-ITT440_${nama_singkatan}_q1.c"
  cd ..
  folder_name="${nama_singkatan}_q2"
  mkdir "$folder_name"
  cd "$folder_name"
  echo "// Name: $nama" > "2503-ITT440_${nama_singkatan}_q2_server.c"
  echo "// Student ID: $nombor_pelajar" >> "2503-ITT440_${nama_singkatan}_q2_server.c"
  echo "// Shortname: $nama_singkatan" >> "2503-ITT440_${nama_singkatan}_q2_server.c"
  echo "// Date: $timestamp" >> "2503-ITT440_${nama_singkatan}_q2_server.c"
  echo "// Q2 (C SERVER) 2503-ITT440 FA" >> "2503-ITT440_${nama_singkatan}_q2_server.c"
  echo "// Name: $nama" > "2503-ITT440_${nama_singkatan}_q2_client.c"
  echo "// Student ID: $nombor_pelajar" >> "2503-ITT440_${nama_singkatan}_q2_client.c"
  echo "// Shortname: $nama_singkatan" >> "2503-ITT440_${nama_singkatan}_q2_client.c"
  echo "// Date: $timestamp" >> "2503-ITT440_${nama_singkatan}_q2_client.c"
  echo "// Q2 (C CLIENT) 2503-ITT440 FA" >> "2503-ITT440_${nama_singkatan}_q2_client.c"
  echo "Maklumat telah disimpan dalam fail 2503-ITT440_${nama_singkatan}_q2_server.c"
  echo "Maklumat telah disimpan dalam fail 2503-ITT440_${nama_singkatan}_q2_client.c"
  cd ..
  folder_name="${nama_singkatan}_q3"
  mkdir "$folder_name"
  cd "$folder_name"
  echo "# Name: $nama" > "2503-ITT440_${nama_singkatan}_q3_server.py"
  echo "# Student ID: $nombor_pelajar" >> "2503-ITT440_${nama_singkatan}_q3_server.py"
  echo "# Shortname: $nama_singkatan" >> "2503-ITT440_${nama_singkatan}_q3_server.py"
  echo "# Date: $timestamp" >> "2503-ITT440_${nama_singkatan}_q3_server.py"
  echo "# Q3 (Py SERVER) 2503-ITT440 FA" >> "2503-ITT440_${nama_singkatan}_q3_server.py"
  echo "# Name: $nama" > "2503-ITT440_${nama_singkatan}_q3_client.py"
  echo "# Student ID: $nombor_pelajar" >> "2503-ITT440_${nama_singkatan}_q3_client.py"
  echo "# Shortname: $nama_singkatan" >> "2503-ITT440_${nama_singkatan}_q3_client.py"
  echo "# Date: $timestamp" >> "2503-ITT440_${nama_singkatan}_q3_client.py"
  echo "# Q3 (Py CLIENT) 2503-ITT440 FA" >> "2503-ITT440_${nama_singkatan}_q3_client.py"
  echo "Maklumat telah disimpan dalam fail 2503-ITT440_${nama_singkatan}_q3_server.py"
  echo "Maklumat telah disimpan dalam fail 2503-ITT440_${nama_singkatan}_q3_client.py"
  cd ..
  folder_name="${nama_singkatan}_q4"
  mkdir "$folder_name"
  cd "$folder_name"
  echo "# Name: $nama" > "2503-ITT440_${nama_singkatan}_q4.py"
  echo "# Student ID: $nombor_pelajar" >> "2503-ITT440_${nama_singkatan}_q4.py"
```

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```
  echo "# Shortname: $nama_singkatan" >> "2503-ITT440_${nama_singkatan}_q4.py"
  echo "# Date: $timestamp" >> "2503-ITT440_${nama_singkatan}_q4.py"
  echo "# Q4 (PARALLEL) 2503-ITT440 FA" >> "2503-ITT440_${nama_singkatan}_q4.py"
  echo "Maklumat telah disimpan dalam fail 2503-ITT440_${nama_singkatan}_q4.py"
  cd ../..
 ;;
 2)
# Section 2
  folder_name="${nama_singkatan}_q1"
  cd 2503-ITT440_Final_Assessment/
  cd "$folder_name"
  gcc -o "${nama_singkatan}" "2503-ITT440_${nama_singkatan}_q1.c"
  #cat "2503-ITT440_${nama_singkatan}_q1.c"
  echo "Kompilasi selesai. Fail output: ${nama_singkatan}"
  cd ../..
 ;;
 3)
# Section 3
  folder_name="${nama_singkatan}_q1"
  cd 2503-ITT440_Final_Assessment/
  cd "$folder_name"
  #cp "2503-ITT440_${nama_singkatan}_q1.c" "${nama_singkatan}"
  if [ -f "${nama_singkatan}" ]; then
   ./"${nama_singkatan}"
   #cat "${nama_singkatan}"
   echo "Program ${nama_singkatan} telah dijalankan."
  else
   echo "Fail output ${nama_singkatan} tidak wujud."
  fi
  cd ../..
 ;;
 4)
# Exit
  echo "Keluar dari skrip."
  break
 ;;
 *)
  echo "Pilihan tidak sah."
 ;;
 esac
 echo "Adakah anda ingin kembali ke menu utama atau keluar?"
 echo "1. Kembali ke menu utama"
 echo "2. Keluar"
 read -p "Masukkan pilihan anda (1/2): " pilihan_ulang
 if [ "$pilihan_ulang" -ne 1 ]; then
  echo "Keluar dari skrip."
  break
 fi
done
tree 2503-ITT440_Final_Assessment
```

Figure 1: Bourne Again Shell Script

(25 marks)

a) Run the script from Figure 1, select ‘Section 1’ and ‘Exit’ then capture the output screen. The script copy is also provided on GitHub (addff/**2503-ITT440**) under Final Assessment.

b) Write a program in C that will fork into 7 child processes, where the parent will display 'We stand with Gaza' every two seconds and each child will display 'Free Palestine' every second. When the user presses Ctrl+C, the parent will display 'Stop Killing! Stop Genocide! Stop Occupation!'. Pass any message between the parent and child processes through a pipe and then display it. The program will exit after 10 seconds. All 15 child processes will wait for the parent to exit first.

c) Provide a screenshot or multiple screenshots of the running program that you wrote. You can use ‘Section 2’ and ‘Section 3’ from the script shown in Figure 1.

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QUESTION 3

You need to develop a multi-user TCP client-server program in C designed to calculate students' CGPA.

(25 marks)

a) Write a server program. The server component must operate on port 20000 or above. When a client connects, it should first identify the current user's name to the server. Following this, the client will send subject details, including the subject code, subject name, its associated credit hours, the grade point and the semester in which it was taken. Upon receiving this data, the server's core function will be to calculate and return the GPA for the current semester, GPA for all previous semesters, and the overall cumulative CGPA for that specific user. If you need a database or file to store records, it is entirely up to your own creativity.

b) Write a client program in C that complies with the transaction of the above server.

c) Please perform as many users and semesters as possible on the terminal and capture the screen while the program is running.

QUESTION 3

You are required to create a client-server program in Python for a chat server. You may use the TCP transmission protocol.

(25 marks)

a) Create a TCP server using any number of ports number in the range 10000 to 10999 The server must register the chat user. The server will pass messages between clients.

b) Create a TCP client. The client must identify the chat user first before chatting. Then the client will ask if the person to chat with is available. If available, the client can send messages through the server. The chat ends if the user types 'keluar!'.

c) Provide multiple screenshots showing the running chat server and client.

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QUESTION 4

Create a multiprocessing program in Python to generate Sudoku questions and answers. The questions must be divided into four difficulties: Easy, Medium, Hard, and Extreme. Please ensure that the programme uses parallel processing with multithreading. If you need a database or file to store generated data, it is entirely up to your own creativity.

(25 marks)

a) Write a program that performs the following tasks in separate processes:

i) Generate and print 100 Extreme Sudoku puzzles.
ii) Generate and print 1000 Hard Sudoku puzzles.
iii) Generate and print 10 000 Intermediate Sudoku puzzles.
iv) Generate and print 100 000 Easy Sudoku puzzles.

b) Provide multiple screenshots showing the execution of the Python program using multiprocessing and multithreading for parallel processing.

END OF QUESTION PAPER

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